

SOFTWARE ENGINEERING ASSIGNMENT REPORT

AI-Powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers

A. Assignment Information

Particular	Details
Department	AIML
Programme	B.Tech – Engineering
Course Code & Course Name	Software Engineering
Academic Year / Batch	2026 Batch
Faculty Name	Dr. M. Arivukarasi
Assignment Title	AI-Powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers
Date of Issue	
Date of Submission	
Maximum Marks	100
Course Outcome(s) – CO	CO1, CO2, CO3, CO4, CO5, CO6
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
SDG Mapping (SDG 1–17)	SDG 9 (Industry, Innovation & Infrastructure), SDG 12 (Responsible Consumption & Production)
Industry / Societal Relevance	Standards compliance and consumer protection for MSMEs, startups, students, and consumers (Ministry of Consumer Affairs / BIS)

B. Assignment Problem / Challenge

The assignment requires the design and development of an AI-powered conversational assistant proposed under the Ministry of Consumer Affairs, Food & Public Distribution / Department of Consumer Affairs (DoCA), Category: Software, Theme: Smart Automation. The Bureau of Indian Standards (BIS) publishes thousands of Indian Standards and offers services such as product certification, hallmarking, laboratory recognition, Standards Clubs, training, and conformity assessment. Users — particularly MSMEs, startups, students, and consumers — currently struggle to identify applicable standards, certification requirements, licensing procedures, and testing requirements because this information is scattered across multiple documents, portals, and PDFs. This assignment applies software engineering principles to build an intelligent, source-backed assistant that resolves this information-retrieval gap through natural-language interaction.

C. Problem Statement

Problem / Case / Design Challenge

Design and develop an AI-powered Intelligent Assistant (Agent) that enables users to obtain accurate, context-aware, and source-backed information related to Indian Standards and BIS services through natural-language interaction. The assistant must understand plain-language queries, retrieve relevant information from authorized BIS knowledge sources, and provide responses that cite the specific documents or clauses applicable — answering questions on Indian Standards, recommending applicable standards from product descriptions, explaining BIS certification schemes and processes, addressing consumer queries, guiding users on hallmarking, and suggesting relevant testing laboratories.

D. Requirements and Constraints

Functional Requirements

- Accept natural-language queries about Indian Standards and BIS services.
- Recommend applicable Indian Standards based on a free-text product description.
- Explain BIS certification schemes, licensing procedures, and step-by-step certification processes.
- Answer general consumer-protection and consumer-affairs queries.
- Provide guidance on hallmarking requirements and procedures for jewellery/precious metals.
- Suggest relevant BIS-recognized testing laboratories based on product category and location.
- Return responses with citations to the specific standard, clause, or scheme document referenced.

Non-Functional / Technical Constraints

- Accuracy: responses must be grounded in authorized BIS documents to avoid hallucinated or non-authoritative answers.
- Performance: query response latency should remain low for an interactive chat experience.
- Scalability: system must scale to handle concurrent queries from MSMEs, students, and consumers nationwide.
- Security & privacy: user queries and any uploaded product details must be handled securely.
- Data currency: the knowledge base must be kept up to date as BIS standards are revised, withdrawn, or newly published.
- Traceability: every answer must be explainable, with a verifiable reference back to the source document/clause.
- Accessibility: interface should be usable by non-technical users, including regional-language support where feasible.

- Cost and resources: solution should be practical to deploy using open, reusable NLP/retrieval components.

E. Proposed Work

1. Problem Understanding and Formulation (CO1, CO2)

What is the problem: BIS knowledge — standards documents, certification schemes, hallmarking rules, and laboratory listings — is fragmented across numerous PDFs and portals, making it difficult and time-consuming, especially for MSMEs, startups, students, and consumers, to find applicable and correctly cited information.

Expected outcomes: A single conversational assistant that answers natural-language questions with accurate, source-backed, and traceable references to the relevant Indian Standard, clause, or BIS scheme document, reducing the time and expertise needed to navigate BIS resources.

Information/data available: the full corpus of published Indian Standards, BIS certification scheme documents, hallmarking guidelines, laboratory recognition lists, and consumer-affairs FAQs/circulars.

Assumptions made: (a) machine-readable versions of BIS standards and scheme documents can be obtained/digitized for indexing; (b) the knowledge corpus can be periodically refreshed as standards are revised; (c) a retrieval-augmented generation (RAG) approach can ground an LLM's responses in this corpus with acceptable accuracy.

Constraints to satisfy: response accuracy and traceability, real-time interactive performance, and secure handling of user data (Section D).

This stage supports CO1 by framing the problem as an agile, iteration-based project (indexing corpus → building retrieval → building conversational layer) and CO2 by performing requirements analysis prior to architecture design.

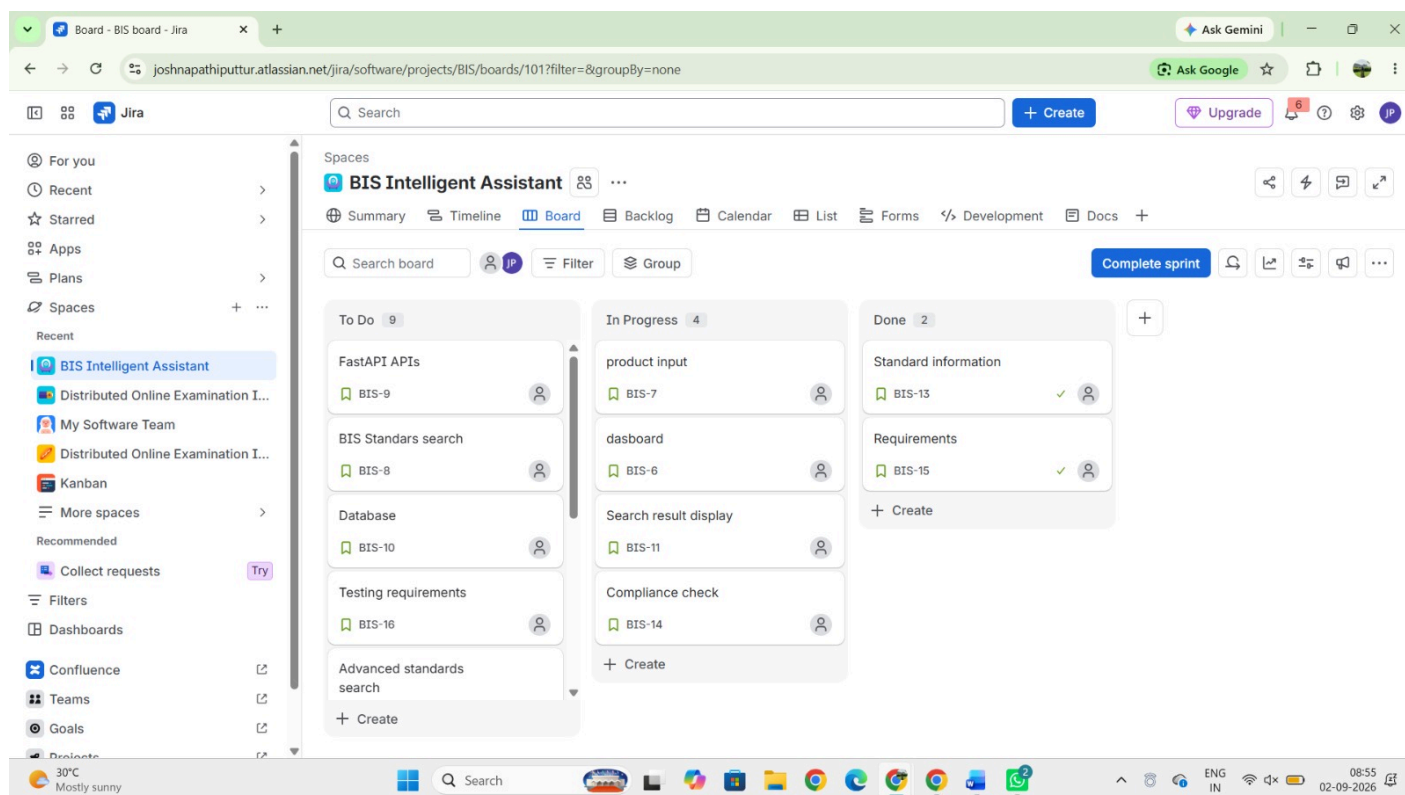
2. Application of Course Knowledge (CO1, CO2)

Agile project management (CO1): the project is organized into Scrum sprints — corpus digitization and indexing, retrieval-engine development, conversational/NLU layer, citation-and-traceability layer, and laboratory-recommendation module — with a prioritized backlog and iterative sprint reviews.

Requirements analysis and prioritization (CO2): requirements (Section D) are prioritized using MoSCoW — accurate, cited answers to standards/certification questions are 'Must have'; laboratory recommendation and hallmarking guidance are 'Should have'; regional-language support is 'Could have'.

Software engineering principles applied: separation of concerns between the knowledge-ingestion pipeline, the retrieval layer, and the conversational-generation layer; API-first design so the assistant can be embedded into web, mobile, or existing BIS portals.

Design methodology: a Retrieval-Augmented Generation (RAG) architecture is adopted, where an LLM is grounded in a continuously updated vector index of BIS documents rather than relying on unconstrained generation — directly addressing the requirement for source-backed, citation-carrying answers.



3. Solution / Design / Methodology (CO2, CO3)

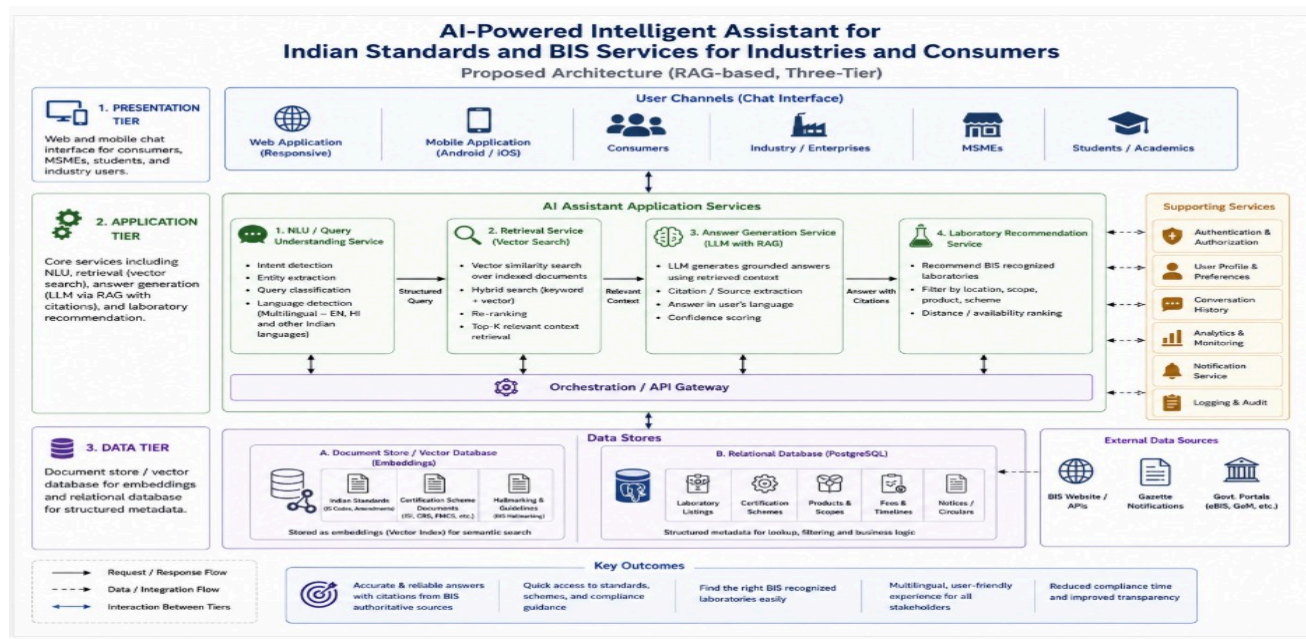
Proposed architecture (RAG-based, three-tier):

- Presentation tier: web and mobile chat interface for consumers, MSMEs, students, and industry users.
- Application tier: an NLU/query-understanding service, a retrieval service (vector search over indexed BIS documents), an answer-generation service (LLM grounded via RAG with citation extraction), and a laboratory-recommendation service.
- Data tier: a document store/vector database holding embeddings of Indian Standards, certification scheme documents, and hallmarking guidelines, plus a relational database (e.g., PostgreSQL) for structured metadata such as laboratory listings and scheme details.

Data flow: BIS standards, scheme documents, and lab-recognition lists are ingested through a document-processing pipeline (OCR/parsing → chunking → embedding) into the vector store; a user's natural-language query is embedded and matched against this store to retrieve the most relevant clauses; the LLM composes a response strictly grounded in the retrieved passages and attaches the corresponding document/clause citation; a separate rules-based/lookup service matches product category and location to recommend BIS-recognized testing laboratories.

Alternative solutions considered: (1) a keyword/full-text search portal — simple to build but cannot answer natural-language or product-description-based queries; (2) an ungrounded general-purpose LLM chatbot — conversational but prone to fabricating standards or clauses, which is unacceptable for a compliance-sensitive domain; (3) the selected RAG-based, citation-grounded architecture, which was selected because it balances natural-language usability with the traceability and accuracy required for standards guidance.

This design stage satisfies CO2 (scalable, requirement-driven architecture) and CO3 (full-stack build with containerized, version-controlled, collaboratively developed microservices).



4. Use of Modern Tools (CO3, CO4)

The following modern tools and techniques are applicable to this solution:

- Backend: Python/FastAPI or Node.js for the retrieval and conversational microservices.
- AI/LLM layer: an LLM (e.g., OpenAI/Llama/Gemini-class model) used within a Retrieval-Augmented Generation pipeline for grounded, citation-carrying responses.
- Document processing: OCR and PDF-parsing tools to digitize and chunk BIS standards and scheme documents.
- Vector search/embedding store for semantic retrieval of relevant standards and clauses.
- Databases: PostgreSQL/MongoDB for structured metadata such as laboratory listings, scheme catalogues, and user query logs.
- Containerization and orchestration: Docker for packaging each microservice, Kubernetes for scaling during peak usage (e.g., MSME registration periods).
- Version control and collaboration: Git/GitHub with feature branches and pull-request-based collaborative development.

- **Github link:**
- CI/CD and DevOps: automated pipelines for building, testing (including retrieval-accuracy regression tests), and deploying updates whenever the BIS knowledge corpus changes.

Students should provide evidence of tool usage — repository history, sample retrieval/embedding outputs, Docker build logs, and pipeline run screenshots — as supporting appendices.

The screenshot shows a VS Code editor with a file explorer on the left, a code editor in the center, and a chat sidebar on the right.

File Explorer (Left):

- SE Project
 - __pycache__
 - app.cpython-311.pyc
 - .venv
 - data
 - bis_assistant.db
 - static
 - css
 - style.css
 - js
 - app.js
 - templates
 - index.html
 - .gitignore
 - app.py (selected)
 - Dockerfile
 - README.md
 - requirements.txt

Code Editor (Center):

```

1  """
2  BIS Assist - Intelligent Assistant for Indian Standards & BIS S
3  Backend: FastAPI + SQLite (Deterministic Rule-Based Engine, No
4  """
5
6  import os
7  import re
8  import sqlite3
9  import datetime
10 from typing import List, Optional, Dict, Any
11 from pathlib import Path
12 from contextlib import asynccontextmanager
13
14 from fastapi import FastAPI, HTTPException, Request, Query
15 from fastapi.responses import HTMLResponse, JSONResponse, FileR
16 from fastapi.staticfiles import StaticFiles
17 from fastapi.middleware.cors import CORSMiddleware
18 from pydantic import BaseModel, Field
19
20 # Base Directory Setup
21 BASE_DIR = Path(__file__).resolve().parent
22 DATA_DIR = BASE_DIR / "data"
23 STATIC_DIR = BASE_DIR / "static"
24 TEMPLATES_DIR = BASE_DIR / "templates"
25 DB_PATH = DATA_DIR / "bis_assistant.db"
26
27 # Ensure data and static directories exist
28 DATA_DIR.mkdir(parents=True, exist_ok=True)
29 STATIC_DIR.mkdir(parents=True, exist_ok=True)
30 (STATIC_DIR / "css").mkdir(parents=True, exist_ok=True)
31 (STATIC_DIR / "js").mkdir(parents=True, exist_ok=True)
32 TEMPLATES_DIR.mkdir(parents=True, exist_ok=True)
  
```

Chat Sidebar (Right):

Chat

Build with Agent

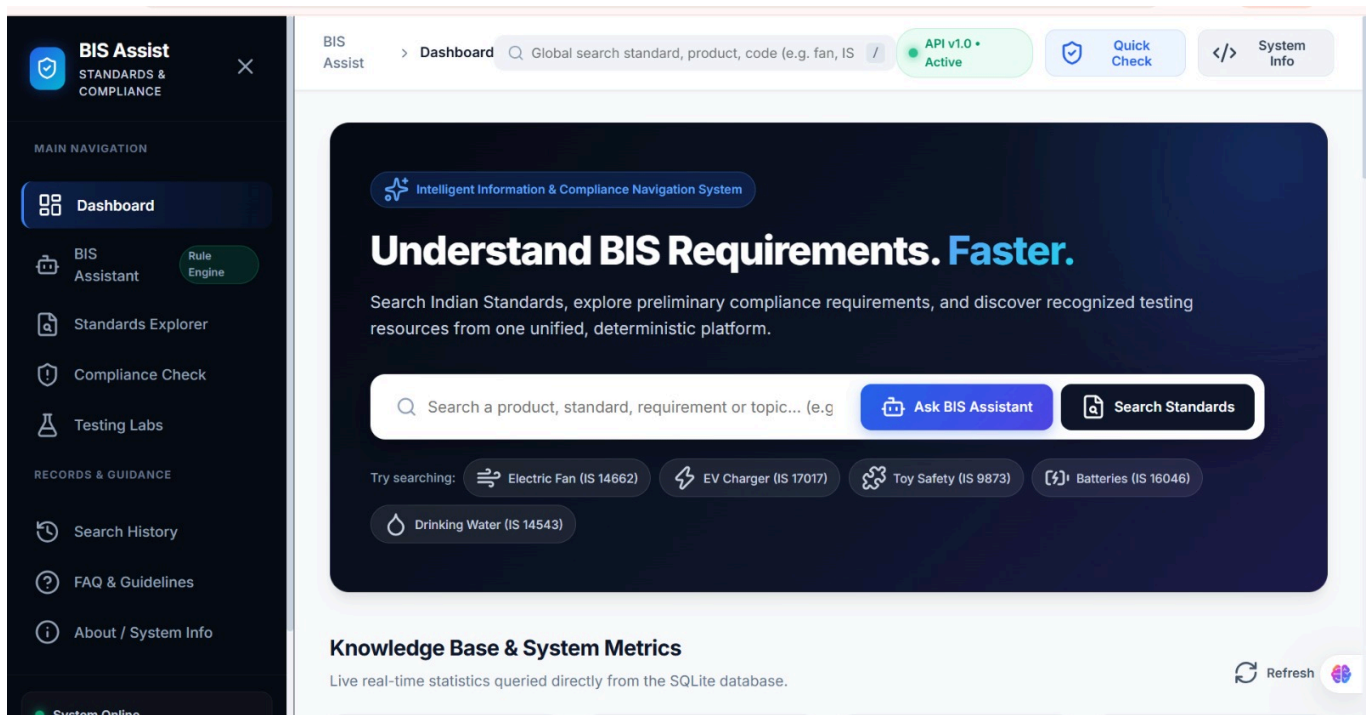
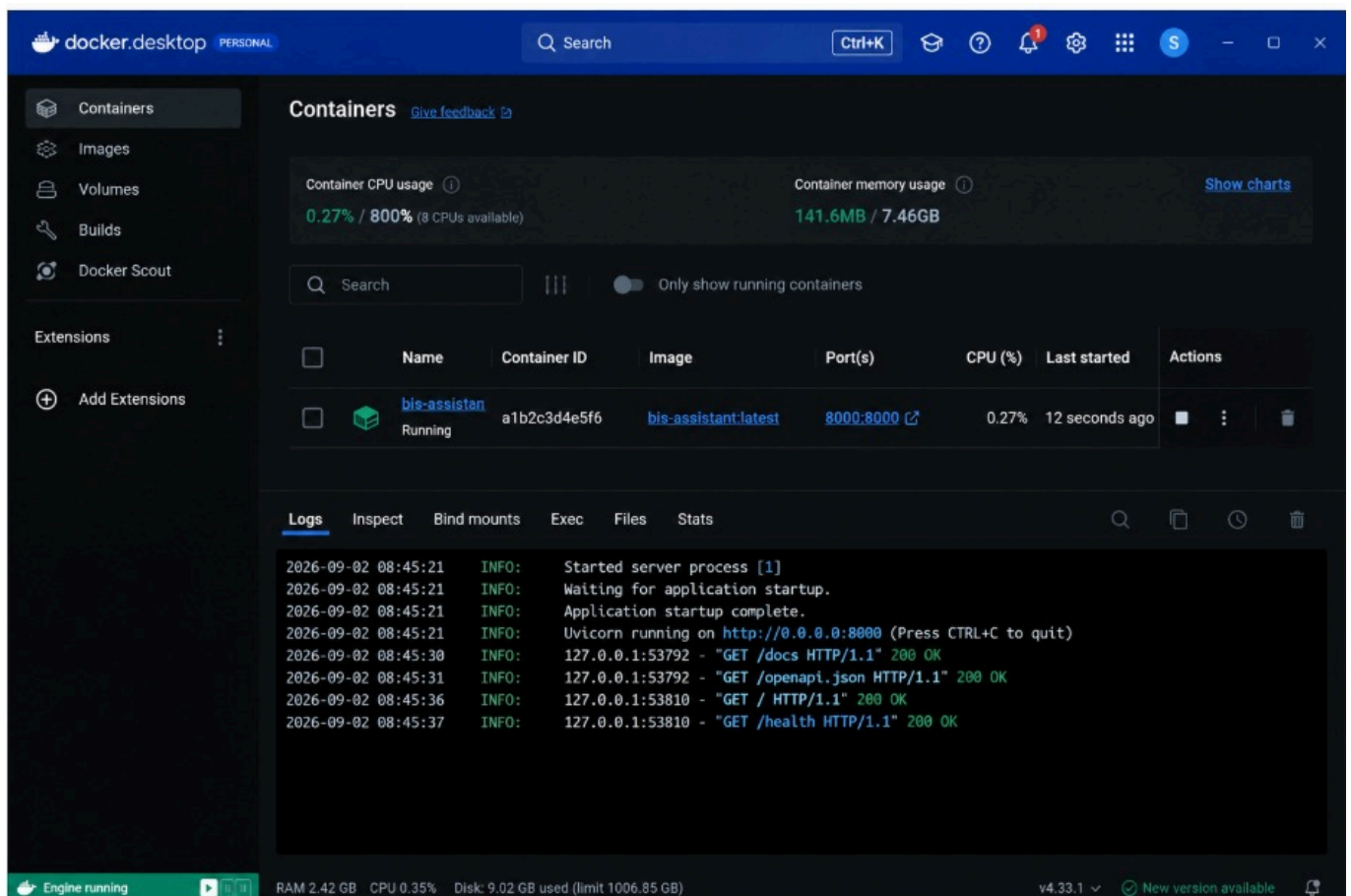
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.

+ app.py

Describe what to build

+ Agent Models

Local Default permissions



5. Results and Validation (CO3, CO4)

Representative validation activities and expected evidence for this assignment include:

- Functional testing: sample queries (e.g., a product description) verified against the correct recommended Indian Standard and its citation.

- Retrieval accuracy testing: precision/recall of the retrieval layer against a manually curated set of question–clause pairs.
- Citation-correctness testing: verification that every generated answer references a genuine, correctly matched document/clause rather than a fabricated one.
- Performance testing: response-latency measurement under concurrent user load.
- Security testing: confirmation that API endpoints are authenticated and user query data is handled securely.
- DevOps validation: successful automated build–test–deploy run, with the knowledge index rebuilt and redeployed whenever source documents are updated.

Metrics to report: retrieval precision/recall, citation accuracy rate, average response latency, and system uptime. Results should be presented using tables, graphs, and screenshots of sample query-response pairs, and validated against the functional and non-functional requirements in Section D.

```
I thought for 94s >
Edited test_selenium.py +6 -4
Ran & "c:\Users\Sailaja Rani\Documents\SE Project\.venv\Scripts\python.exe" test_selenium.py >
Explored 1 task >
Run full Selenium test suite finished >
Project\BIS_Assist_Selenium_Test_Report.xlsx

=====
===== TEST EXECUTION COMPLETED:
12/12 PASSED (100.0%) in 55.76s EXCEL REPORT: C:\Users\Sailaja
Rani\Documents\SE Project\BIS_Assist_Selenium_Test_Report.xlsx

Working
```

6. Analysis and Engineering Decision (CO2, CO4, CO5)

Comparing the RAG-based approach against the keyword-search and ungrounded-LLM alternatives (Section E.3): the RAG architecture offers substantially higher factual reliability and traceability, since every answer is anchored to a retrieved source passage, at the cost of additional engineering effort to build and maintain the document-ingestion and indexing pipeline.

Trade-off analysis: grounding responses strictly in retrieved BIS text trades some conversational flexibility (the assistant may decline to answer when no matching clause is found) for much higher trustworthiness — an

appropriate trade-off in a compliance and certification context where an incorrect answer could mislead a business's regulatory decisions.

The final solution — a RAG-based, citation-grounded, containerized conversational assistant — is justified quantitatively by its retrieval precision/recall and citation-accuracy metrics, and qualitatively by its suitability for a domain where traceability to an authoritative source is non-negotiable.

7. Broader Considerations (CO5)

- Economics: reduces the time and consultancy cost MSMEs and startups currently spend interpreting standards and certification requirements.
- Society & accessibility: makes standards and certification information accessible to non-expert consumers and students, not just industry compliance specialists.
- Professional responsibility & ethics: because the domain is regulatory/compliance-sensitive, the assistant must clearly indicate when it cannot find a grounded answer rather than guessing, to avoid misinforming users about legal or safety requirements.
- Consumer protection: accurate hallmarking and certification guidance directly supports consumer trust and protection against substandard or mislabelled products, aligning with the Department of Consumer Affairs' mandate.
- Sustainability: encourages responsible production and consumption (SDG 12) by making compliance with quality and safety standards easier for producers to achieve.
- Data governance: since the corpus includes regulatory documents, version control and audit trails of which standard revision an answer was based on are important for accountability.

8. Conclusion (CO5, CO6)

This assignment proposed an AI-powered Intelligent Assistant for Indian Standards and BIS Services that consolidates fragmented BIS knowledge — standards, certification schemes, hallmarking rules, and laboratory listings — into a single, natural-language, citation-grounded conversational interface. Using a Retrieval-Augmented Generation architecture with a continuously updated vector index, containerized microservices, and a CI/CD pipeline, the proposed solution satisfies the key functional requirements of accurate standard recommendation, certification guidance, and lab suggestion, while meeting non-functional requirements of accuracy, traceability, scalability, and security.

Major findings: grounding LLM responses in a retrieval layer over authoritative BIS documents is essential for trustworthiness in a compliance-sensitive domain, and this trade-off is justified despite added engineering complexity. Limitations: the solution depends on the availability and quality of digitized/machine-readable BIS documents, and keeping the knowledge base current requires an ongoing update pipeline as standards are revised. Possible improvements: adding regional-language support, integrating direct links to BIS's official licensing portals for seamless action-taking, and incorporating user feedback to continuously refine retrieval quality.

Overall, the assignment synthesizes and applies concepts from across the course — agile management, requirements and architecture, full-stack and containerized implementation, testing and DevOps, and evaluation of societal/professional impact — as a comprehensive capstone-style exercise, directly addressing CO6.

9. Student Reflection (CO5, CO6)

What I learned beyond the classroom: This problem showed how, in a regulatory-compliance domain, correctness and traceability of an AI system's answers matter as much as its conversational fluency — which reframed my understanding of 'good' AI-assistant design away from pure natural-language quality and toward grounded, verifiable retrieval.

With additional time/resources: Given more time and resources, I would build a working prototype of the retrieval pipeline over a sample subset of publicly available Indian Standards, measure actual retrieval precision/recall against a hand-labelled query set, and test the citation-accuracy of generated answers before extending the assistant to the full BIS corpus.

10. References

- [1] Bureau of Indian Standards (BIS), Ministry of Consumer Affairs, Food & Public Distribution — official standards, certification, and hallmarking documentation.
- [2] Department of Consumer Affairs (DoCA) circulars and consumer-guidance publications.
- [3] Literature on Retrieval-Augmented Generation (RAG) architectures for grounded question answering.
- [4] Docker and Kubernetes official documentation, for containerization and orchestration references.
- [5] Vector database and embedding-model documentation as used in the proposed retrieval layer.
- [6] AI-assisted tools: an AI language model was used to assist in structuring and drafting portions of this report, in accordance with the assignment's requirement to acknowledge AI-assisted tools used.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO1, CO2	PO1, PO2	L3/L4	10
Application of knowledge	CO1, CO2	PO1, PO2	L3/L4	20
Solution / Design / Implementation	CO2, CO3	PO3, PO5	L4/L5/L6	20
Modern tool usage	CO3, CO4	PO5	L3/L4	10
Validation	CO3, CO4	PO4, PO5	L4/L5	15
Analysis & justification	CO4, CO5	PO4	L4/L5	15
Broader considerations	CO5	PO6, PO8	L4/L5	5
Documentation & reflection	CO6	PO9, PO10	L3/L4	5

Note: Only the POs and Bloom's levels genuinely assessed by this assignment have been mapped; not every component maps to every PO.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment:

Actual CO Attainment:

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering: